Measurement-Fueled Conditional Bridges in Graph-State Echo Experiments

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Abstract

We report numerical observations from a Qiskit-based simulation study of measurement-fueled conditional bridges in a five-qubit graph-state echo experiment. A mid-circuit measurement of an ancilla qubit is used as "fuel" to conditionally apply a controlled-Z (CZ) gate between a designated qubit pair (bridge) in the middle of a Loschmidt echo sequence. We observe that this conditional bridge consistently suppresses echo return probability to zero in the $m = 1$ measurement branch at scrambler depths $T = 2$ and $T = 4$, while the $m = 0$ branch retains partial return. We further observe that the disruptive strength of the bridge depends on the topological relationship between the bridge pair and the brickwork scrambler structure: pairs overlapping with scrambler edges show stronger disruption than pairs orthogonal to the scrambler pattern. These are preliminary numerical results on an ideal, noise-free simulator. Physical interpretation and scalability beyond five qubits remain open questions.

1 Introduction

Quantum error correction and near-term quantum computing share a common challenge: decoherence and information leakage are not fully preventable, but may be partially controllable through careful circuit design. One approach is to treat mid-circuit measurements not as passive readout operations but as active resources—using classical measurement outcomes to steer subsequent quantum operations.

This work explores a specific instance of this idea: a mid-circuit measurement result is used to decide whether a CZ gate, here called a "bridge", is applied between two qubits that are not directly connected in the base graph state. The experiment is framed as a Loschmidt echo, which provides a clean operational definition of return: how well does the system recover its initial state after scrambling and attempted unscrambling?

The core question is whether the conditional bridge, applied only in the $m = 1$ measurement branch, affects the echo return, and whether its topological position in the graph matters.

We do not claim new physics. We report reproducible numerical observations and propose them as a basis for further investigation.

2 Experimental Setup

2.1 Graph State

The initial state is a five-qubit star graph state with qubit 0 as hub and edge set

$$E = \{(0,1),(0,2),(0,3),(0,4)\}. \tag{1}$$

It is prepared as

$$|G\rangle = CZ_{04}CZ_{03}CZ_{02}CZ_{01}|+\rangle^{\otimes 5}. \tag{2}$$

The graph-state stabilizers

$$K_v = X_v \bigotimes_{u\in N(v)} Z_u \tag{3}$$

satisfy $\langle K_v\rangle = +1$ for the unperturbed state, providing a natural quality metric for echo return.

2.2 Scrambler $U(T)$

The scrambler consists of T layers. Each layer applies Hadamard gates to all qubits, followed by CZ gates on brickwork pairs. Even layers use $(0,1)$ and $(2,3)$, while odd layers use $(1,2)$ and $(3,4)$.

The inverse $U^\dagger(T)$ applies the same layers in reverse order. Since H and CZ are self-inverse, this inverse is exact. In the absence of any perturbation or bridge,

$$U^\dagger(T)U(T) = I, \tag{4}$$

and the system returns perfectly to $|G\rangle$.

2.3 Perturbation W

A Z gate is applied to qubit 0 between $U(T)$ and the measurement step. This perturbation breaks the perfect echo symmetry and is the signal whose scrambling we track.

2.4 Measurement-Fueled Bridge Protocol

The full circuit sequence is

$$\begin{aligned} &|G\rangle \to U(T) \to Z(q_0) \to H(q_1) \to \text{measure}(q_1) \to m \\ &\quad \to \begin{cases} CZ(\text{bridge pair}), & m=1, \\ I, & m=0, \end{cases} \to U^\dagger(T) \to \text{final measurement}. \end{aligned} \tag{5}$$

The rule qubit q_1 is placed in superposition by H before measurement, ensuring an approximately equal probability for $m=0$ and $m=1$. A roughly 50/50 split is observed in all runs, confirming no systematic bias.

The bridge CZ is applied only when $m=1$. This creates two distinct branches of the experiment within a single run: the $m=0$ branch, in which no bridge is applied and $U^\dagger(T)$ attempts the standard echo, and the $m=1$ branch, in which a bridge CZ is applied before $U^\dagger(T)$ and unscrambling is attempted against a modified circuit. Results are postselected on m and analyzed separately.

Table 1: Baseline echo: mean $\langle K \rangle$ and per-qubit stabilizer values.

T	Mean $\langle K \rangle$	K_0	K_1	K_2	K_3	K_4
0	+0.60	−1.00	+1.00	+1.00	+1.00	+1.00
1	−0.60	+1.00	−1.00	−1.00	−1.00	−1.00
2	**+1.00**	+1.00	+1.00	+1.00	+1.00	+1.00
3	−0.60	+1.00	−1.00	−1.00	−1.00	−1.00
4	+0.60	+1.00	+1.00	−1.00	+1.00	+1.00
6	+0.20	+1.00	+1.00	−1.00	+1.00	−1.00
8	+0.20	−1.00	+1.00	+1.00	+1.00	−1.00

2.5 Observables

Three observables are computed per branch:

- $P(\text{return})$: the probability that all qubits are measured in $|0\rangle$, used as an echo return proxy;
- $\langle Z_k \rangle$: a single-qubit Z expectation for $k = \text{bridge_pair}[0]$;
- $\langle Z_k Z_l \rangle$: a two-qubit ZZ correlator for $(k, l) = \text{bridge_pair}$.

The baseline echo without bridge and without shots uses the ideal `StatevectorEstimator` to compute stabilizer expectations $\langle K_v \rangle$ as a reference.

2.6 Simulation Parameters

All simulations use the Qiskit Aer `AerSimulator` in an ideal, noise-free setting. Each circuit is sampled with 4096 shots. Transpilation uses a five-qubit line coupling map and the basis gates `rz`, `sx`, `x`, `cx`, and `cz`. No physical QPU runs have been performed at this stage.

3 Results

3.1 Baseline Echo

Table 1 shows stabilizer expectations for the baseline echo, with a Z perturbation on qubit 0 and no bridge, across scrambler depths T.

The periodic pattern, in which $T = 2$ gives perfect stabilizer return with mean $+1$, reflects the symmetry of the brickwork scrambler on this topology. At $T = 2$, $U^\dagger(T)Z(q_0)U(T)$ produces a net operation that maps all stabilizers back to $+1$, suggesting that the Z perturbation is effectively absorbed into the echo at this depth.

3.2 Bridge Experiment: Main Result

Table 2 shows $P(\text{return})$ and $\langle Z_0 \rangle$ for bridge pair $(0, 4)$ across all tested values of T, postselected on m.

The primary observation is that at $T = 0, 2, 4, 8$—the depths where the baseline echo shows partial or full stabilizer return—$P_{\text{ret}}|m1$ drops to exactly 0.000, while $P_{\text{ret}}|m0$ retains values near 0.12–0.13. The bridge, applied only in the $m = 1$ branch, completely eliminates echo return at these depths.

Table 2: Bridge pair $(0,4)$: $P(\text{return})$ and $\langle Z_0 \rangle$ by measurement branch.

T	$n(m=0)$	$P_{\text{ret}}\|m0$	$\langle Z_0 \rangle\|m0$	$n(m=1)$	$P_{\text{ret}}\|m1$	$\langle Z_0 \rangle\|m1$
0	~ 2050	0.123	**+1.000**	~ 2050	**0.000**	**−1.000**
1	~ 2020	0.000	−1.000	~ 2080	0.034	−0.008
2	~ 2080	0.132	+0.012	~ 2020	**0.000**	+0.005
3	~ 2060	0.033	−0.023	~ 2040	0.031	−0.011
4	~ 2040	0.126	−0.013	~ 2050	**0.000**	−0.032
6	~ 2040	0.029	−0.020	~ 2050	0.030	+0.006
8	~ 2160	0.117	**+1.000**	~ 1930	**0.000**	**−1.000**

Table 3: Bridge topology sweep: mean $|\Delta P_{\text{ret}}|$ ranked by disruption.

Bridge pair	In base graph	Mean $\|\Delta P_{\text{ret}}\|$	Scrambler overlap
$(0,4)$	yes (star edge)	0.087	partial (q_4 in even layers)
$(1,2)$	no	0.086	high (both in odd layer)
$(0,1)$	yes (star edge)	0.083	high (both in even layer)
$(2,3)$	no	0.072	high (both in even layer)
$(1,3)$	no	**0.046**	low (no shared layer)

The secondary observation is that at $T=0$ and $T=8$, $\langle Z_0 \rangle$ is sharply polarized: $\langle Z_0 \rangle|m0 = +1.000$ and $\langle Z_0 \rangle|m1 = -1.000$. This is a deterministic flip of qubit 0's Z polarization conditioned on the bridge measurement. The $CZ(0,4)$ bridge, applied after scrambling, produces a clean Z eigenstate inversion on the hub qubit when the echo would otherwise return it to a definite state.

At $T=6$, the two branches converge, with $P_{\text{ret}}|m0 \approx P_{\text{ret}}|m1 \approx 0.03$, suggesting that at higher scrambler depths the bridge's effect is absorbed by the increased complexity of the unscrambling task.

3.3 Bridge Topology Sweep

Table 3 shows the mean disruption $|\Delta P_{\text{ret}}| = |P_{\text{ret}}(m=0) - P_{\text{ret}}(m=1)|$ averaged over $T \in \{2,4,6\}$ for five bridge pairs.

The pair $(1,3)$ is consistently the least disruptive across all tested T values. Pairs $(0,4)$, $(1,2)$, $(0,1)$, and $(2,3)$ cluster in a higher disruption band.

The distinguishing structural property is scrambler overlap: pairs $(1,2)$ and $(2,3)$ are even/odd brickwork edges, and $(0,1)$ is both a star edge and a brickwork edge. The pair $(1,3)$ shares no brickwork layer with the scrambler at any parity.

A notable anomaly occurs at $T=4$: bridge pairs $(1,2)$ and $(1,3)$ both produce $\langle Z_0 \rangle|m0 = +1.000$ and $\langle Z_0 \rangle|m1 = -1.000$, the same sharp Z flip seen for bridge $(0,4)$ at $T=0$ and $T=8$. This suggests that the Z-flip phenomenon is not unique to hub-touching bridges, but depends on the interplay between bridge position and scrambler depth.

4 Discussion

4.1 What the Data Supports

The measurements support the following empirical claims. First, a mid-circuit measurement used to conditionally apply a CZ bridge between a qubit pair significantly affects echo return at scrambler depths $T = 2$ and $T = 4$ on a five-qubit star graph state. Second, the effect is asymmetric: $m = 0$ retains $P_{\text{ret}} \approx 0.12$–$0.13$, while $m = 1$ drops to $P_{\text{ret}} = 0.000$ at these depths. Third, the pair $(1, 3)$, whose qubits do not share any brickwork layer, shows approximately half the mean disruption of other tested pairs. Fourth, at select bridge/depth combinations, $\langle Z_k \rangle$ is sharply polarized, with opposite signs in the $m = 0$ and $m = 1$ branches.

4.2 What the Data Does Not Support

The following claims would require additional evidence not present in this work. Scalability is unknown because all results are on five qubits. Physical relevance is also untested because all simulations use an ideal, noise-free model, and the observed effects may be modified or obscured by realistic noise. We observe correlation between scrambler edge overlap and disruption strength, but do not provide a formal proof that this overlap is the causal mechanism. Finally, the results are specific to the brickwork scrambler on a star graph; different scrambler designs or graph topologies may produce different behavior.

4.3 Potential Interpretations

The suppression of P_{ret} to zero in the $m = 1$ branch at $T = 2$ and $T = 4$ can be understood informally as follows. The inverse $U^\dagger(T)$ is the exact inverse of $U(T)$ only when no additional operations are inserted between them. The bridge CZ introduces an extra unitary operation that is not part of the scrambler structure. Consequently, $U^\dagger(T)$ unscrambles the wrong circuit—one that includes an extra CZ—and the result is not $|G\rangle$.

The question of why $(1, 3)$ is less disruptive than other pairs is more subtle. A possible explanation is that the brickwork scrambler propagates information along its own edge structure. A bridge that shares an edge with the scrambler inserts its effect directly into the scrambler's communication channels, while a bridge orthogonal to the scrambler inserts its effect in a direction that the scrambler does not efficiently mix.

This is a hypothesis, not a result. Verifying it would require a formal analysis of the scrambler's operator spreading properties, which is beyond the scope of this note.

5 Open Questions

The following directions are natural next steps:

1. **Noise model validation.** Run the same circuits on Aer with a realistic noise model, such as depolarizing noise plus T_1/T_2 relaxation. Does the bridge effect survive at moderate noise levels?

2. **Scaling.** Repeat on seven- and nine-qubit systems. Does the $(1, 3)$-type topology, in which the bridge is orthogonal to all scrambler layers, remain the weakest disruptor?

3. **Scrambler independence.** Test with a random unitary scrambler instead of brickwork CZ. If topology dependence disappears, it is a scrambler artifact; if it persists, it may be a graph property.

4. **Formal channel analysis.** Express the bridge as a quantum channel acting on the echo circuit and derive its effect analytically. This requires density matrix formalism beyond the scope of the current code.

5. **QPU validation.** Run the $T = 2$ case on IBM Quantum hardware. The circuit is shallow enough, with depth roughly 15 after transpilation, to be within reach of current NISQ devices.

6 Conclusion

We have presented numerical observations from a five-qubit graph-state echo experiment in which a mid-circuit measurement conditionally applies a CZ bridge between qubit pairs. The bridge consistently suppresses echo return probability to zero in the $m = 1$ branch at scrambler depths $T = 2$ and $T = 4$, while $m = 0$ retains partial return. A topology sweep across five bridge pairs suggests that pairs overlapping with scrambler brickwork edges are more disruptive than pairs orthogonal to the scrambler structure.

These are preliminary results from ideal simulation on five qubits. They motivate further investigation into the relationship between graph topology, scrambler structure, and measurement-fueled conditional operations in near-term quantum circuits.

Code is available at QHDALabs/qmnet.

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Simulation code: QHDALabs/qmnet—`qmnet_v3.py`.
Data: all results are reproducible by running `qmnet_v3.py` with default parameters.